

Review — Read Before You Start

Quick Reference

Polygons: interior angle sum of an n -gon is $(n - 2) \cdot 180^\circ$. Regular n -gon has each interior angle $\frac{(n - 2)180^\circ}{n}$.

Triangles: interior sum = 180° . Equilateral: all sides equal, each angle 60° . Isosceles: base angles equal.

Parallel lines and transversals: corresponding angles are equal; alternate interior angles are equal; interior same-side are supplementary.

Triangle similarity & proportional segments: if triangles are similar, corresponding sides are proportional; transversals cutting parallel lines create proportional segments.

Exponents: $a^m a^n = a^{m+n}$; $\frac{a^m}{a^n} = a^{m-n}$ ($a \neq 0$); $(a^m)^n = a^{mn}$; $(ab)^m = a^m b^m$; $a^{-k} = \frac{1}{a^k}$; $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$.

Algebra tips: combine like terms; distribute carefully; watch restrictions (denominator $\neq 0$); keep exact forms when possible.

Test Mindset

Sketch small diagrams, label what is given, and note what is parallel or equal. For exponents, rewrite with positive exponents last. For word problems, define the variable clearly.

Part A — Geometry

1. A polygon has interior angle sum 1980° . How many sides does it have?

.....

2. The interior angle of a regular polygon is 168° . How many sides does the polygon have?

.....

3. In $\triangle ABC$, $\angle A = 3x + 10^\circ$, $\angle B = 5x - 22^\circ$, and $\angle C = 2x + 8^\circ$. Find x and the three

angles.

.....
.....
.....
.....

4. Lines $\ell_1 \parallel \ell_2$. A transversal creates corresponding angles of measures $(2y + 14)^\circ$ and $(5y - 52)^\circ$. Solve for y and give the common angle measure.

.....
.....
.....
.....

5. Two transversals cut parallel lines forming segments on one transversal of lengths 6 cm and 9 cm, and on the other transversal of 8 cm and x cm, in the same order from top to bottom. Find x .

.....
.....
.....

6. In $\triangle PQR$, $QR \parallel ST$ with S on PQ and T on PR . If $PS : SQ = 2 : 3$ and $PR = 25$, find PT .

.....
.....
.....

7. $\triangle XYZ$ is equilateral. The perimeter is 30 cm. A segment joins a vertex to the midpoint of the opposite side. Find the length of that segment.

.....
.....
.....

8. A triangle has side lengths in the ratio 3 : 4 : 5 and perimeter 60 cm. The largest angle

is opposite which side, and what is its measure?

.....

.....

.....

.....

9. A regular hexagon and an equilateral triangle share the same side length s . Compare their perimeters and areas (express each in terms of s and $\sqrt{3}$).

.....

.....

.....

.....

.....

10. A triangle has two angles measuring x and $2x + 10$. The exterior angle at the third vertex is 140° . Find the three interior angles.

.....

.....

.....

.....

Part B — Exponents & Algebra

11. Write with positive exponents only: $\frac{x^{-3}y^2}{xy^{-5}}$.

.....

.....

.....

12. Simplify: $(3a^2b^{-1}) \cdot \left(\frac{2a^{-3}b^4}{6a}\right)$.

.....

.....

.....

13. Simplify: $\left(\frac{2x^3y^{-2}}{4x^{-1}y}\right)^{-2}$.

.....

.....

.....

.....

14. Evaluate exactly: $\left(\frac{1}{2}\right)^{-3} \cdot 2^{-1}$.

.....

.....

.....

15. Simplify completely: $\frac{12x^4y^{-2}}{18x^{-1}y^5} \cdot \left(\frac{3x^2y}{2}\right)^{-1}$.

.....

.....

.....

.....

16. Combine like terms and simplify: $4x^2 - 3x + 7 - 2x^2 + 5x - 9$.

.....

.....

17. Move the factor with a negative exponent to make exponents positive: $\frac{5a^{-2}b^3}{10a b^{-4}}$ (no decimals).

.....

.....

.....

18. Distribute and simplify: $3(2x - 5) - 2(4x + 1)$.

.....

.....

19. Solve for x : $\frac{x-3}{4} + \frac{x+1}{6} = \frac{5}{12}$.

.....

.....

.....

.....

20. Solve for t : $5(2t-1) - 3(t+7) = 4t+8$.

.....

.....

.....

21. Word problem (single variable): A phone plan charges a base fee of \$12 plus \$0.15 per text. If the monthly bill (before tax) is \$39.30, how many texts were sent?

.....

.....

.....

22. Word problem with fractions: A tank is $\frac{3}{5}$ full. After adding 12 gallons, it is $\frac{7}{8}$ full. What is the tank's total capacity (in gallons)?

.....

.....

.....

.....

23. Hard expression — simplify fully with positive exponents:

$$E = \left(\frac{a^{-2}b^3}{a b^{-1}} \right)^2 \cdot \left(\frac{a^3}{b} \right)^{-1} \cdot \frac{b^{-4}}{a^{-1}}.$$

.....

.....

.....

.....

.....

24. Simplify: $\frac{(x^{1/3}y^{-2/3})^6}{x^2y^{-1}}$.

.....
.....
.....
.....

25. Evaluate and simplify: $\left(\frac{9m^{-2}}{27m}\right)^{-1} \cdot (3m^0)^{-2}$.

.....
.....
.....
.....

Show units where appropriate, and leave exact forms with π or radicals unless the question asks for a decimal.